并行计算实验 1

接入实验环境

登录进入网关:

```
Last login: Thu Apr 23 14:13:55 2026 from 192.168.0.131
renjianxin@cccourse-gate:~ $
renjianxin@cccourse-gate:~ $
```

登录主节点 raspi00

```
renjianxin@cccourse-gate:~ $
renjianxin@cccourse-gate:~ $ ssh raspi00
renjianxin@raspi00's password:
Linux raspi00 4.19.66-v7+ #1253 SMP Thu Aug 15 11:49:46 BST 2019 arm
v7l

The programs included with the Debian GNU/Linux system are free soft
ware;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Apr 23 13:27:31 2026 from 192.168.86.9
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
```

准备工作

生成密钥对

```
renjianxin@raspi00:~/example $ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/renjianxin/.ssh/id_rsa):
Created directory '/home/renjianxin/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/renjianxin/.ssh/id_rsa.
Your public key has been saved in /home/renjianxin/.ssh/id_rsa.pub.
The key fingerprint is:
SHA256:Oftoc+MoU+RuxFu92XfSn2y/qT5YbsOQDk/ORbNLA0E renjianxin@raspi0
0
The key's randomart image is:
+---[RSA 2048]----+
|          .E     |
|           .     |
|            .    |
|        .. . o   |
|       +S  .+ o  |
|        =+.+.*   |
|       +.oB O+o. |
|      o *+o*oB+.*|
|       =o+o.oo=BB|
+----[SHA256]-----+
```

链接 userscrits 文件夹

```
renjianxin@raspi00:~ $ ln -s /home/pi/userscripts
renjianxin@raspi00:~ $ ll
-bash: ll: 未找到命令
renjianxin@raspi00:~ $ ls  /home/pi/userscripts
cluster-cptonodes  cluster-nodes  cluster-rmdir   cluster-sync
cluster-mkdir      cluster-rm     cluster-setkey
renjianxin@raspi00:~ $ chmod 600 pass
chmod: 无法访问'pass': 没有那个文件或目录
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $ ls
example  userscripts
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $ cd userscripts
renjianxin@raspi00:~/userscripts $ ls
cluster-cptonodes  cluster-nodes  cluster-rmdir   cluster-sync
cluster-mkdir      cluster-rm     cluster-setkey
renjianxin@raspi00:~/userscripts $
```

链接 example 文件夹

```
renjianxin@raspi00:~ $ ln -s /home/pi/examples
renjianxin@raspi00:~ $ ls
examples  nodes.list  userscripts
renjianxin@raspi00:~ $
```

raspi00 发送公钥

```
renjianxin@raspi00:~ $ ssh-copy-id raspi00
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home
/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s)
, to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if yo
u are prompted now it is to install the new keys
renjianxin@raspi00's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'raspi00'"
and check to make sure that only the key(s) you wanted were added.
```

```
renjianxin@raspi00:~ $ ssh-copy-id raspi01
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@raspi01's password:
Permission denied, please try again.
renjianxin@raspi01's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'raspi01'"
and check to make sure that only the key(s) you wanted were added.
```

```
renjianxin@raspi00:~ $ ssh-copy-id raspi02
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@raspi02's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'raspi02'"
and check to make sure that only the key(s) you wanted were added.

renjianxin@raspi00:~ $ ssh-copy-id raspi03
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
The authenticity of host 'raspi03 (192.168.86.13)' can't be established.
ECDSA key fingerprint is SHA256:Wv20LaS0GaiGpk4qrC1TJRDcZ9Bg5XiGkQgA7N7eAQk.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@raspi03's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'raspi03'"
and check to make sure that only the key(s) you wanted were added.

renjianxin@raspi00:~ $ ssh-copy-id raspi04
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
The authenticity of host 'raspi04 (192.168.86.14)' can't be established.
ECDSA key fingerprint is SHA256:epIjiEujg5zJTvoanhyEfSZZA3Oid3HwaZepiXHCWIg.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@raspi04's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'raspi04'"
and check to make sure that only the key(s) you wanted were added.
```

```
renjianxin@raspi00:~ $ ssh-copy-id raspi10
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
The authenticity of host 'raspi10 (192.168.86.20)' can't be established.
ECDSA key fingerprint is SHA256:epIjiEujg5zJTvoanhyEfSZZA3Oid3HwaZepiXHCWIg.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@raspi10's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'raspi10'"
and check to make sure that only the key(s) you wanted were added.
```

```
renjianxin@raspi00:~ $ ssh-copy-id raspi11
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
The authenticity of host 'raspi11 (192.168.86.21)' can't be established.
ECDSA key fingerprint is SHA256:epIjiEujg5zJTvoanhyEfSZZA3Oid3HwaZepiXHCWIg.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@raspi11's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'raspi11'"
and check to make sure that only the key(s) you wanted were added.

renjianxin@raspi00:~ $ ssh-copy-id raspi12
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
The authenticity of host 'raspi12 (192.168.86.22)' can't be established.
ECDSA key fingerprint is SHA256:epIjiEujg5zJTvoanhyEfSZZA3Oid3HwaZepiXHCWIg.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@raspi12's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'raspi12'"
and check to make sure that only the key(s) you wanted were added.

renjianxin@raspi00:~ $ ssh-copy-id raspi13
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
The authenticity of host 'raspi13 (192.168.86.23)' can't be established.
ECDSA key fingerprint is SHA256:epIjiEujg5zJTvoanhyEfSZZA3Oid3HwaZepiXHCWIg.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@raspi13's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'raspi13'"
and check to make sure that only the key(s) you wanted were added.
```

```
renjianxin@raspi00:~ $ ssh-copy-id raspi14
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
The authenticity of host 'raspi14 (192.168.86.24)' can't be established.
ECDSA key fingerprint is SHA256:epIjiEujg5zJTvoanhyEfSZZA3Oid3HwaZepiXHCWIg.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@raspi14's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'raspi14'"
and check to make sure that only the key(s) you wanted were added.
```

```
renjianxin@raspi00:~ $ ssh-copy-id jetson15
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
The authenticity of host 'jetson15 (192.168.86.25)' can't be established.
ECDSA key fingerprint is SHA256:6yHoal/xqoec3ZANwW/e3R2l/7C6dtDxHB3q0izp0Mw.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@jetson15's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'jetson15'"
and check to make sure that only the key(s) you wanted were added.

renjianxin@raspi00:~ $ ssh-copy-id jetson05
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
The authenticity of host 'jetson05 (192.168.86.15)' can't be established.
ECDSA key fingerprint is SHA256:/iBO6g4GUcm/2xoPUTCMZLVwCKPrqiTrqPgCZ50QYpw.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
renjianxin@jetson05's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'jetson05'"
and check to make sure that only the key(s) you wanted were added.

renjianxin@raspi00:~ $
```

拷贝节点列表

```
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $ cp /home/pi/nodes.list .
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $ ls -al
总用量 36
drwxr-xr-x   4 renjianxin renjianxin 4096 4月  23 18:54 .
drwxr-xr-x 114 root       root       4096 4月  22 19:40 ..
-rw-------   1 renjianxin renjianxin  301 4月  23 16:20 .bash_history
-rw-r--r--   1 renjianxin renjianxin  220 5月  16  2017 .bash_logout
-rw-r--r--   1 renjianxin renjianxin 3523 4月  18  2018 .bashrc
drwxr-xr-x   2 renjianxin renjianxin 4096 4月  23 15:23 example
-rw-r--r--   1 renjianxin renjianxin   90 4月  23 18:54 nodes.list
-rw-r--r--   1 renjianxin renjianxin  675 5月  16  2017 .profile
drwx------   2 renjianxin renjianxin 4096 4月  23 16:08 .ssh
lrwxrwxrwx   1 renjianxin renjianxin   20 4月  23 15:28 userscripts -> /home/pi/userscr
ipts
renjianxin@raspi00:~ $ cat nodes.list
raspi01
raspi02
raspi03
raspi04
jetson05
raspi10
raspi11
raspi12
raspi13
raspi14
jetson15
renjianxin@raspi00:~ $
```

配置 ssh 免密登陆

```
renjianxin@raspi00:~ $ userscripts/cluster-setkey pass nodes.list
Set public key to raspi01 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

Set public key to raspi02 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

Set public key to raspi03 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

Set public key to raspi04 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

Set public key to jetson05 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

Set public key to raspi10 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

Set public key to raspi11 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
```

```
Set public key to raspi11 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

Set public key to raspi12 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

Set public key to raspi13 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

Set public key to raspi14 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

Set public key to jetson15 ...
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/renjianxin/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
                (if you think this is a mistake, you may want to use -f option)

renjianxin@raspi00:~ $
```

测试访问

```
renjianxin@raspi00:~ $ ssh raspi01
Linux raspi01 4.19.66-v7+ #1253 SMP Thu Aug 15 11:49:46 BST 2019 armv7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
renjianxin@raspi01:~ $ exit
注销
Connection to raspi01 closed.
renjianxin@raspi00:~ $ ssh jetson05
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.9.253-tegra aarch64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

扩展安全维护（ESM）Infrastructure 未启用。

0 更新可以立即应用。

163 个额外的安全更新可以通过 ESM Infra 来获取安装。
可通过以下途径了解如何启用 ESM Infra: for Ubuntu 18.04 at
https://ubuntu.com/18-04

renjianxin@jetson05:~$ exit
logout
Connection to jetson05 closed.
```

编译 helloworld 文件

```
renjianxin@raspi00:~/test $ vim hello_mpi.cpp
renjianxin@raspi00:~/test $
renjianxin@raspi00:~/test $
renjianxin@raspi00:~/test $ cat hello_mpi.cpp
#include <mpi.h>
#include <iostream>
#include <math.h>

int main(int argc, char* argv[])
{
     int myid, numprocs;
     int namelen;
     char processor_name[MPI_MAX_PROCESSOR_NAME] = {0};
     //initialize MPI
        MPI_Init(&argc, &argv);
       // get rank of this process
       MPI_Comm_rank(MPI_COMM_WORLD, &myid);
       // get total number of processes over MPI_COMM_WORLD
       MPI_Comm_size(MPI_COMM_WORLD, &numprocs);
       // get the processor name of this process
       MPI_Get_processor_name(processor_name, &namelen);
       std::cout << "Hello World! Process " << myid << " of " << numprocs << " on " << processor_name << std::endl;
       // finalize MPI (release resources, etc.)
        MPI_Finalize();
        return 0;

}
renjianxin@raspi00:~/test $ mpic++ -o hello_mpi hello_mpi.cpp
renjianxin@raspi00:~/test $
```

群发 helloworld 文件

```
renjianxin@raspi00:~ $ cp test/hello_mpi hello_mpi
renjianxin@raspi00:~ $ ls
examples  hello_mpi  nodes.list  test  userscripts
renjianxin@raspi00:~ $
renjianxin@raspi00:~ $ userscripts/cluster-cptonodes hello_mpi nodes.list
Copy file hello_mpi to raspi01 ...
hello_mpi                                              100%   19KB   5.9MB/s   00:00
Copy file hello_mpi to raspi02 ...
hello_mpi                                              100%   19KB   4.5MB/s   00:00
Copy file hello_mpi to raspi03 ...
hello_mpi                                              100%   19KB   6.7MB/s   00:00
Copy file hello_mpi to raspi04 ...
hello_mpi                                              100%   19KB   4.7MB/s   00:00
Copy file hello_mpi to jetson05 ...
hello_mpi                                              100%   19KB   8.4MB/s   00:00
Copy file hello_mpi to raspi10 ...
hello_mpi                                              100%   19KB   4.4MB/s   00:00
Copy file hello_mpi to raspi11 ...
hello_mpi                                              100%   19KB   6.5MB/s   00:00
Copy file hello_mpi to raspi12 ...
hello_mpi                                              100%   19KB   4.4MB/s   00:00
Copy file hello_mpi to raspi13 ...
hello_mpi                                              100%   19KB   6.2MB/s   00:00
Copy file hello_mpi to raspi14 ...
hello_mpi                                              100%   19KB   4.5MB/s   00:00
Copy file hello_mpi to jetson15 ...
hello_mpi                                              100%   19KB   5.9MB/s   00:00
renjianxin@raspi00:~ $
```

修改节点

```
renjianxin@raspi00:~ $ cp nodes.list raspi_hosts.list
renjianxin@raspi00:~ $ vim raspi_hosts.list
renjianxin@raspi00:~ $ cat raspi_hosts.list
raspi00
raspi01
raspi02
raspi03
raspi04
raspi10
raspi11
raspi12
raspi13
raspi14
renjianxin@raspi00:~ $
```

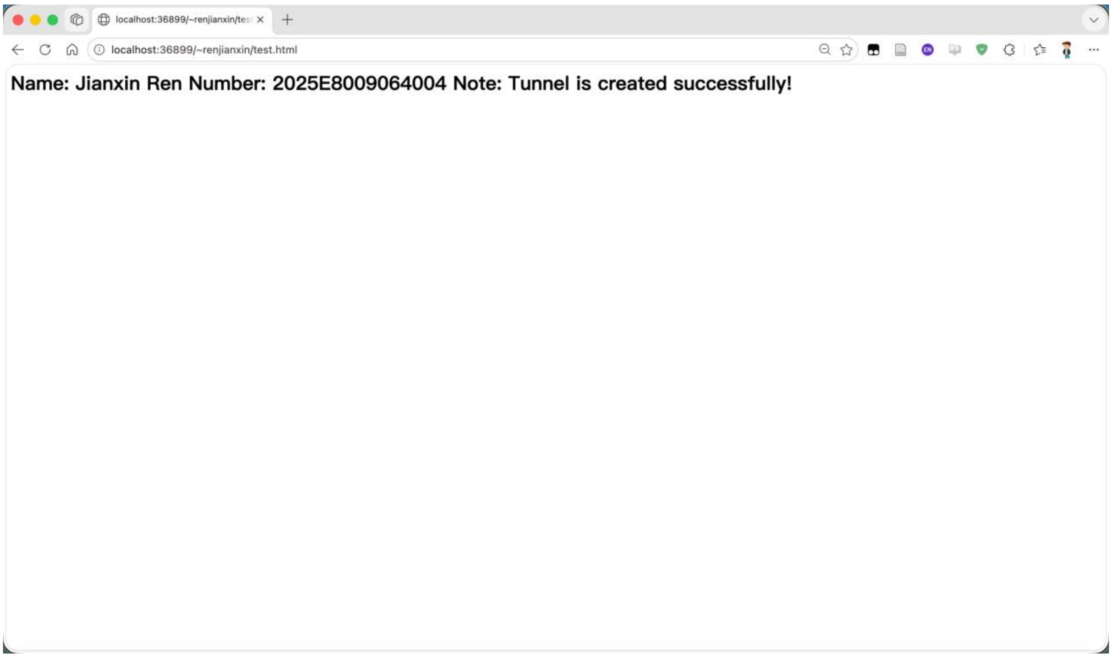
运行测试环境

```
renjianxin@raspi00:~ $ mpirun -np 40 -hostfile raspi_hosts.list  hello_mpi
Hello World! Process 1 of 40 on raspi00
Hello World! Process 2 of 40 on raspi00
Hello World! Process 3 of 40 on raspi00
Hello World! Process 8 of 40 on raspi02
Hello World! Process 9 of 40 on raspi02
Hello World! Process 10 of 40 on raspi02
Hello World! Process 11 of 40 on raspi02
Hello World! Process 16 of 40 on raspi04
Hello World! Process 17 of 40 on raspi04
Hello World! Process 0 of 40 on raspi00
Hello World! Process 18 of 40 on raspi04
Hello World! Process 12 of 40 on raspi03
Hello World! Process 19 of 40 on raspi04
Hello World! Process 13 of 40 on raspi03
Hello World! Process 14 of 40 on raspi03
Hello World! Process 15 of 40 on raspi03
Hello World! Process 32 of 40 on raspi13
Hello World! Process 33 of 40 on raspi13
Hello World! Process 34 of 40 on raspi13
Hello World! Process 35 of 40 on raspi13
Hello World! Process 4 of 40 on raspi01
Hello World! Process 5 of 40 on raspi01
Hello World! Process 36 of 40 on raspi14
Hello World! Process 6 of 40 on raspi01
Hello World! Process 37 of 40 on raspi14
Hello World! Process 7 of 40 on raspi01
Hello World! Process 38 of 40 on raspi14
Hello World! Process 24 of 40 on raspi11
Hello World! Process 39 of 40 on raspi14
Hello World! Process 25 of 40 on raspi11
Hello World! Process 28 of 40 on raspi12
Hello World! Process 29 of 40 on raspi12
Hello World! Process 26 of 40 on raspi11
Hello World! Process 20 of 40 on raspi10
Hello World! Process 30 of 40 on raspi12
Hello World! Process 27 of 40 on raspi11
Hello World! Process 21 of 40 on raspi10
Hello World! Process 22 of 40 on raspi10
Hello World! Process 23 of 40 on raspi10
Hello World! Process 31 of 40 on raspi12
renjianxin@raspi00:~ $
```

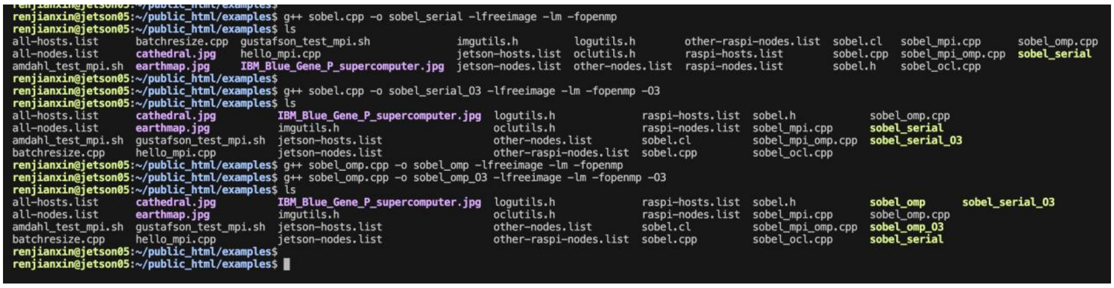
建立隧道链接

```
jachynren — ssh -i ~/Documents/renjianxin -nNT -L 36899:raspi00:368...

[(base) jachynren@JachynRendeMacBook-Air ~ % ssh -i /Users/jachynren/Documents/re]
njianxin -nNT -L 36899:raspi00:36899 -p 31696 renjianxin@310f281v89.qicp.vip
The authenticity of host '[310f281v89.qicp.vip]:31696 ([118.195.239.160]:31696)'
 can't be established.
ED25519 key fingerprint is: SHA256:u9rGSH+etYtCbJZ07aa7w+H45Z8p8HySbf2mcBbf8Lc
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[310f281v89.qicp.vip]:31696' (ED25519) to the list o
f known hosts.
[Enter passphrase for key '/Users/jachynren/Documents/renjianxin':              ]
[Enter passphrase for key '/Users/jachynren/Documents/renjianxin':              ]


123456




[^C%                                                                            ]
(base) jachynren@JachynRendeMacBook-Air ~ % ssh -i /Users/jachynren/Documents/re
njianxin -nNT -L 36899:raspi00:36899 -p 31696 renjianxin@310f281v89.qicp.vip
[Enter passphrase for key '/Users/jachynren/Documents/renjianxin':              ]
```

```
renjianxin@raspi00:~/public_html $
renjianxin@raspi00:~/public_html $ vim test.html
renjianxin@raspi00:~/public_html $ cat test.html
<h1>
Name: Jianxin Ren
Number: 2025E8009064004
Note: Tunnel is created successfully!
</h1>
renjianxin@raspi00:~/public_html $
```



编译文件:



jetson05 建立链接:

```
jachynren — renjianxin@cccourse-gate: ~ — ssh -i ~/Documents/renjian...

Last login: Fri Apr 24 10:34:49 on ttys005
(base) jachynren@JachynRendeMacBook-Air ~ % ssh -i /Users/jachynren/Documents/re
njianxin -L 8888:jetson05:8000 -p 31696 renjianxin@310f281v89.qicp.vip
Enter passphrase for key '/Users/jachynren/Documents/renjianxin':
Last login: Fri Apr 24 11:16:59 2026 from 192.168.0.131
renjianxin@cccourse-gate:~ $
```

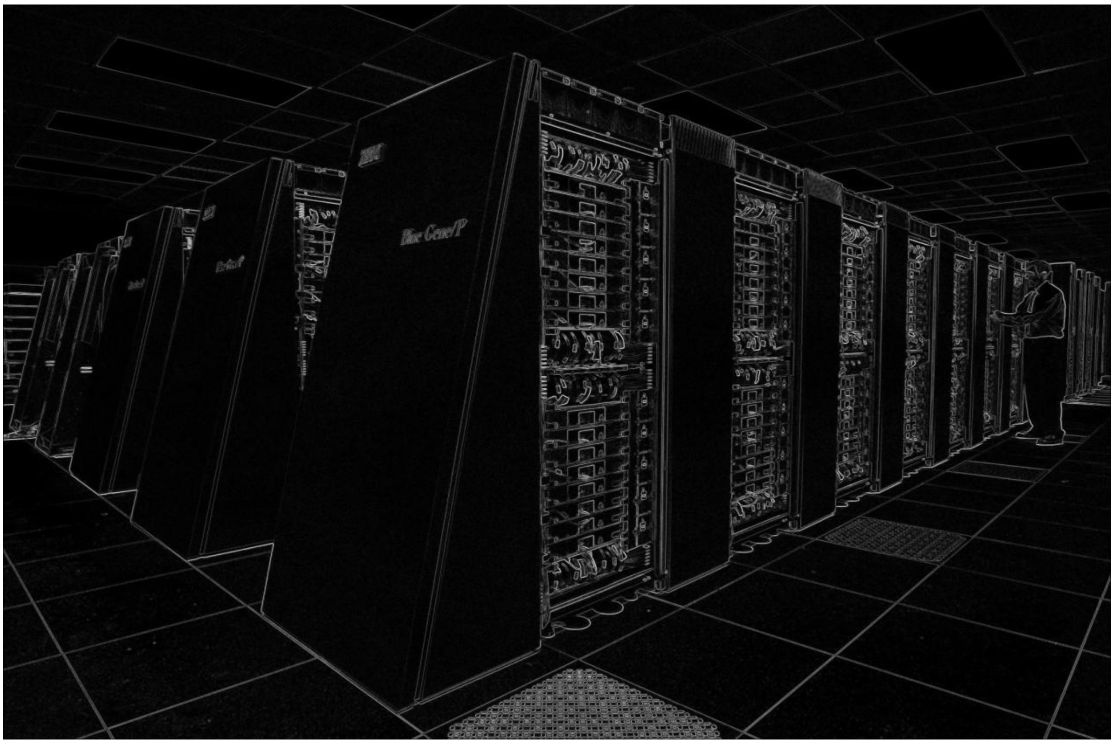
实验结果

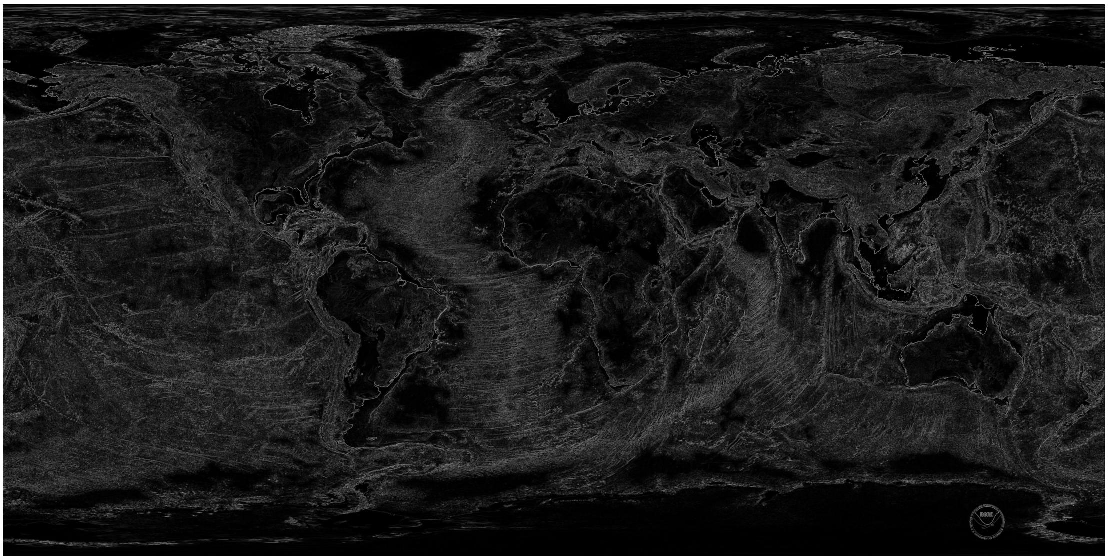
sobel_serial

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_serial IBM_Blue_Gene_P_supercomputer.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 1052.84
The total time for execution is 7.65837s
renjianxin@jetson05:~/public_html/examples$ ls
all-hosts.list      cathedral.jpg          IBM_Blue_Gene_P_supercomputer.jpg      jetson-nodes.list  other-raspi-nodes.list  sobel.cpp          sobel_ocl.cpp  sobel_serial
all-nodes.list      earthmap.jpg           IBM_Blue_Gene_P_supercomputer_out.jpg  logutils.h         raspi-hosts.list        sobel.h            sobel_omp      sobel_serial_O3
amdahl_test_mpi.sh  gustafson_test_mpi.sh  imgutils.h                             oclutils.h         raspi-nodes.list        sobel_mpi.cpp      sobel_omp.cpp
batchresize.cpp     hello_mpi.cpp          jetson-hosts.list                      other-nodes.list   sobel.cl                sobel_mpi_omp.cpp  sobel_omp_O3
renjianxin@jetson05:~/public_html/examples$
```

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_serial earthmap.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 942.731
The total time for execution is 38.3598s
renjianxin@jetson05:~/public_html/examples$
```

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_serial cathedral.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 690.076
The total time for execution is 134.356s
renjianxin@jetson05:~/public_html/examples$
```


Blue Gene/P



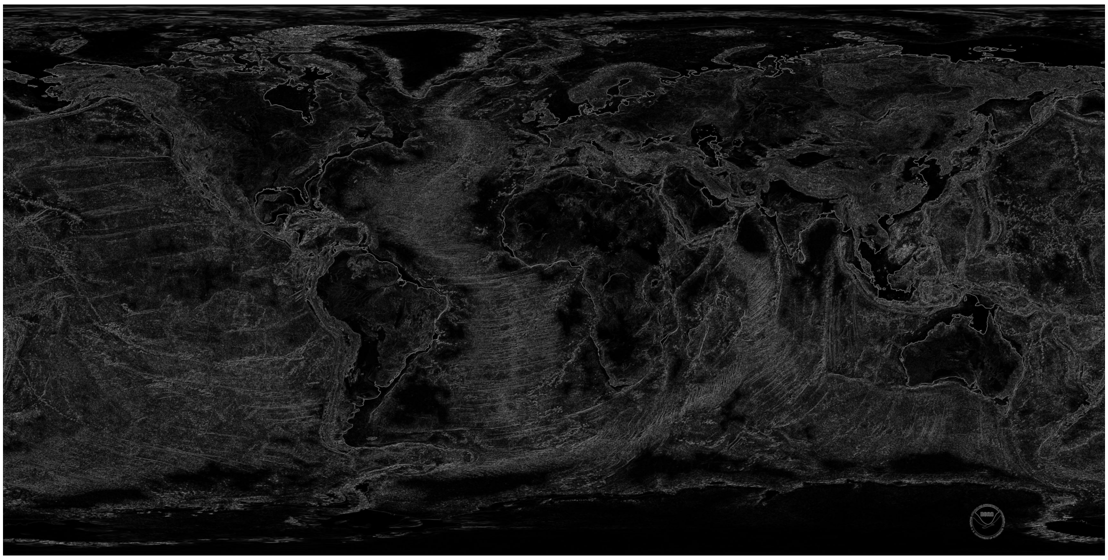
sobel_serial_03

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_serial_03 IBM_Blue_Gene_P_supercomputer.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 1052.84
The total time for execution is 1.89975s
```

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_serial_03 cathedral.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 690.076
The total time for execution is 34.1287s
```

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_serial_03 earthmap.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 942.731
The total time for execution is 9.44979s
```


Blue Gene/P

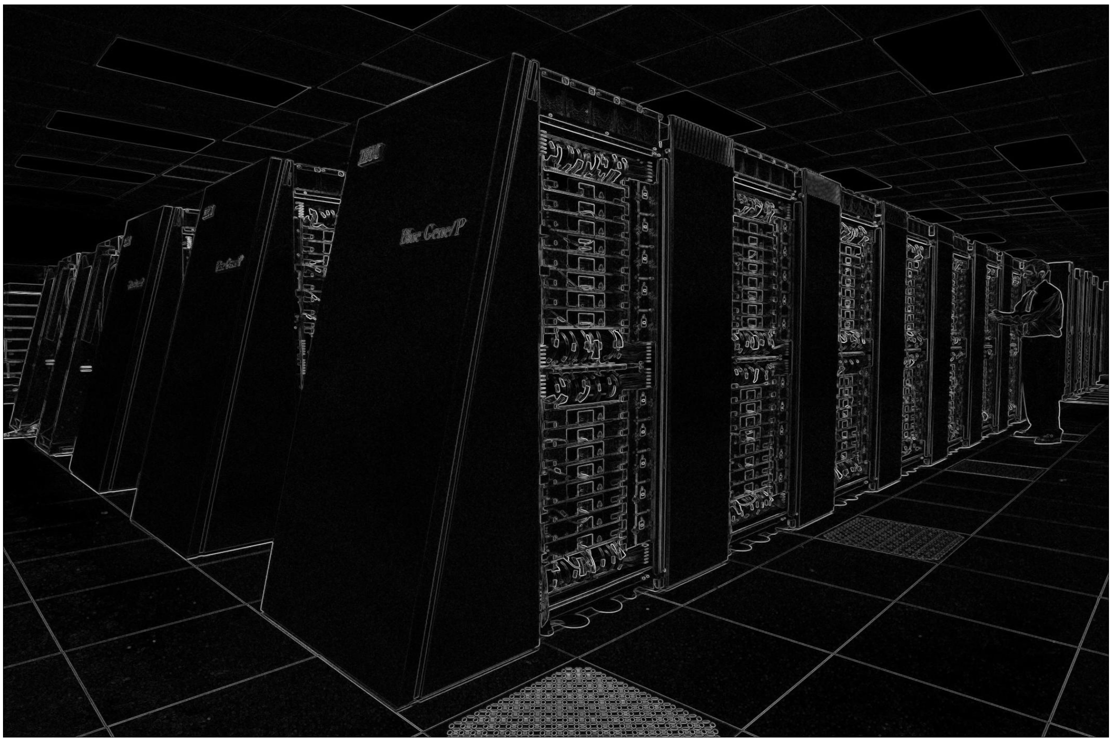


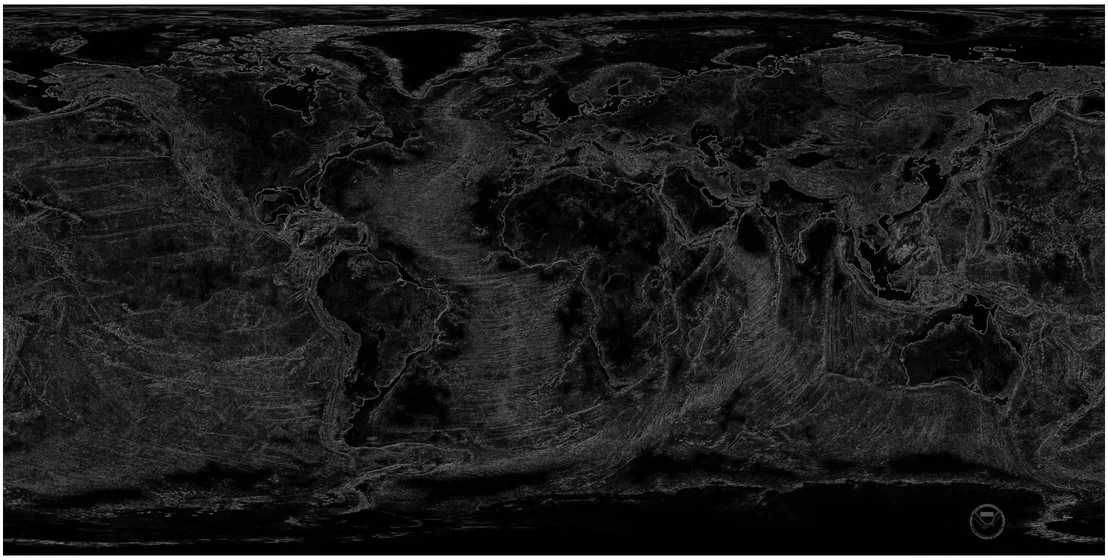
sobel_omp

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_omp IBM_Blue_Gene_P_supercomputer.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 1052.84
The total time for execution is 1.93455s
```

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_omp earthmap.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 942.731
The total time for execution is 9.72089s
```

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_omp cathedral.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 690.076
The total time for execution is 33.7722s
```


Blue Gene/P

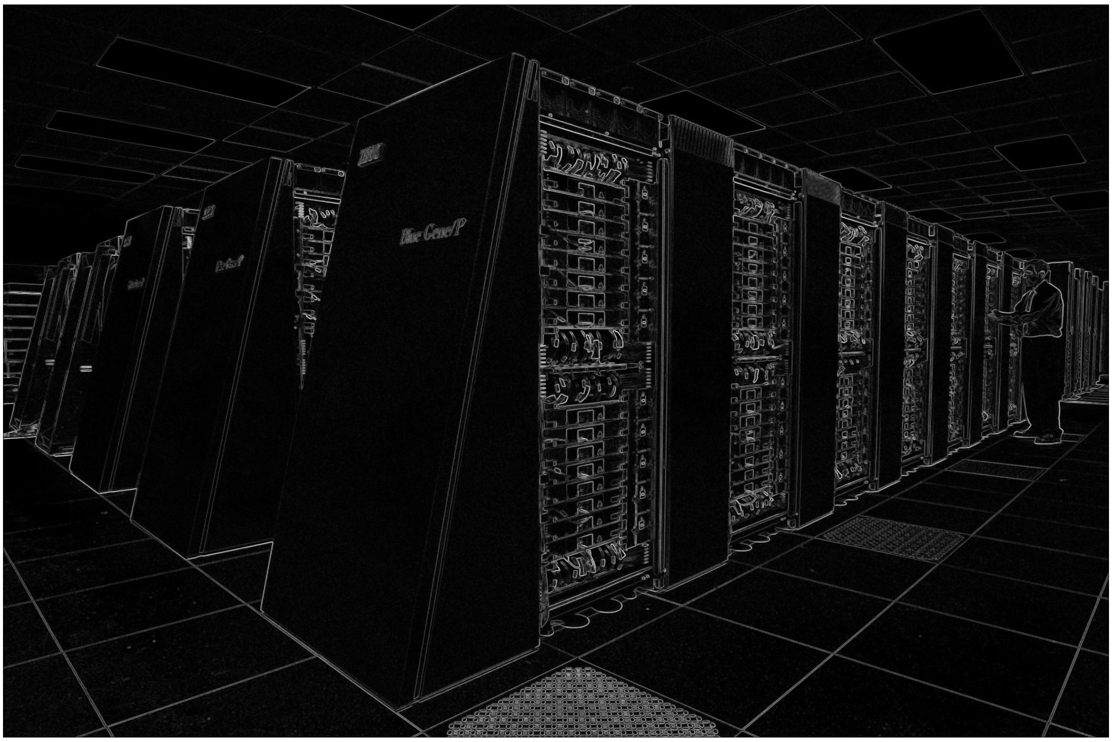


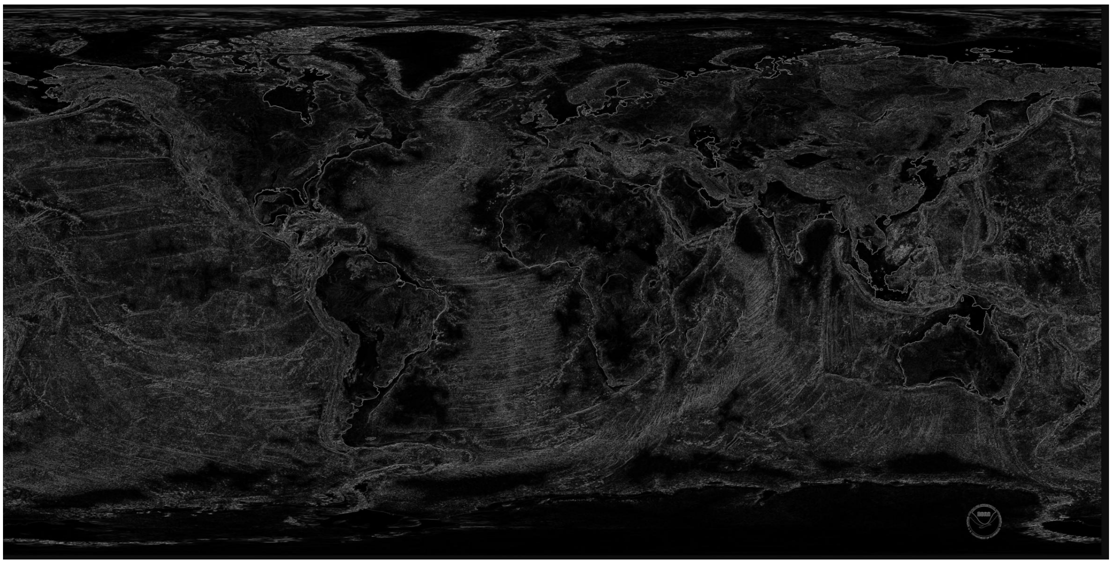
sobel_omp_O3

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_omp_03 IBM_Blue_Gene_P_supercomput
er.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 1052.84
The total time for execution is 0.598822s
```

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_omp_03 earthmap.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 942.731
The total time for execution is 2.9676s
```

```
renjianxin@jetson05:~/public_html/examples$ ./sobel_omp_03 cathedral.jpg
Filtering of input image start ...
the minimum value: 0
the maximum value: 690.076
The total time for execution is 10.3436s
```

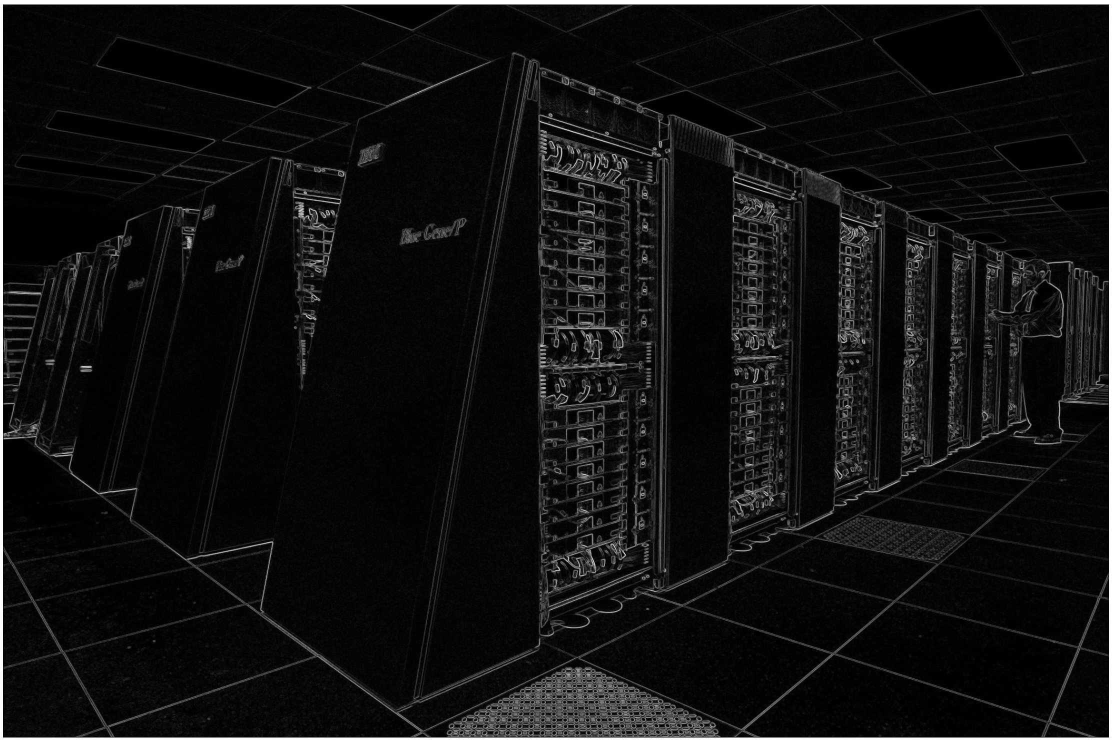

Blue Gene/P

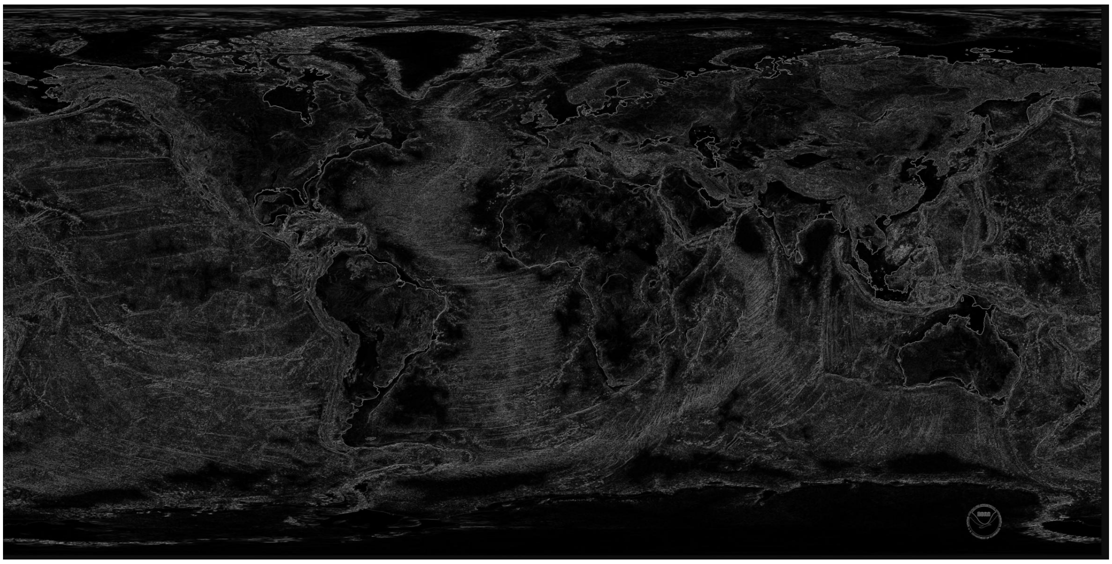


sobel_cuda

```
renjianxin@jetson05:~/public_html/test/examples$ nvcc -o sobel_cuda sobel_cuda.cu -lfreeimage
renjianxin@jetson05:~/public_html/test/examples$ ls
amdahl_test_mpi.sh  earthmap.jpg                    imgutils.h          oclutils.h  sobel_cuda.cu      sobel_mpi_omp.cpp
batchresize.cpp     gustafson_test_mpi.sh           jetson-hosts.list   sobel.cl    sobel.h            sobel_ocl.cpp
cathedral.jpg       hello_mpi.cpp                   jetson-nodes.list   sobel.cpp   sobel_mpi.cpp      sobel_omp.cpp
cudautils.h         IBM_Blue_Gene_P_supercomputer.jpg  logutils.h       sobel_cuda  sobel_mpi_cuda.cu
renjianxin@jetson05:~/public_html/test/examples$
```

```
renjianxin@jetson05:~/public_html/test/examples$ time ./sobel_cuda IBM_Blue_Gene_P_supercomputer.jpg output_cuda.jpg
1 CUDA devices found!
CUDA device 0: NVIDIA Tegra X1
  multi processor count: 1
  shared memory per block: 48 KB
  max threads per block: 1024
  max threads per multi processor: 2048
  max warps per multi processor: 64
Use CUDA device 0
the minimum value: 0
the maximum value: 1059.25
The total time for execution is: 0.171761s

real    0m0.783s
user    0m0.472s
sys     0m0.124s
renjianxin@jetson05:~/public_html/test/examples$ time ./sobel_cuda earthmap.jpg output_cuda1.jpg
1 CUDA devices found!
CUDA device 0: NVIDIA Tegra X1
  multi processor count: 1
  shared memory per block: 48 KB
  max threads per block: 1024
  max threads per multi processor: 2048
  max warps per multi processor: 64
Use CUDA device 0
the minimum value: 0
the maximum value: 962.468
The total time for execution is: 0.411797s

real    0m2.207s
user    0m1.576s
sys     0m0.228s
renjianxin@jetson05:~/public_html/test/examples$ time ./sobel_cuda cathedral.jpg output_cuda2.jpg
1 CUDA devices found!
CUDA device 0: NVIDIA Tegra X1
  multi processor count: 1
  shared memory per block: 48 KB
  max threads per block: 1024
  max threads per multi processor: 2048
  max warps per multi processor: 64
Use CUDA device 0
the minimum value: 0
the maximum value: 999.015
The total time for execution is: 1.03218s

real    0m4.657s
user    0m3.084s
sys     0m0.652s
renjianxin@jetson05:~/public_html/test/examples$
```



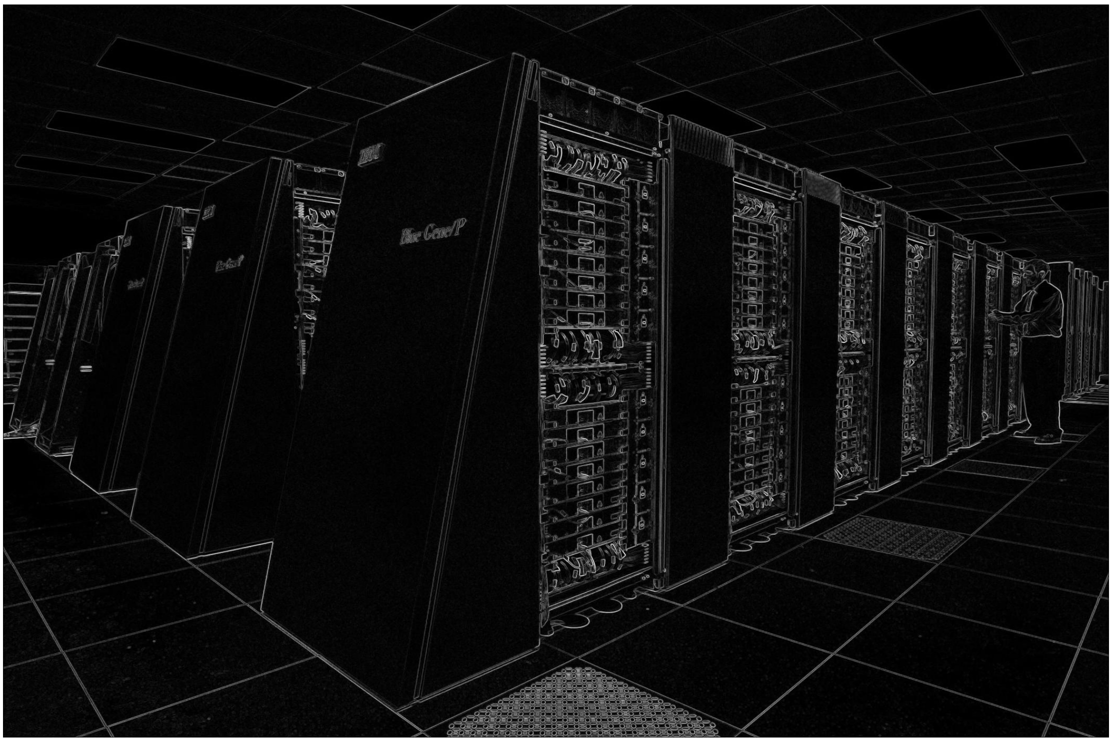


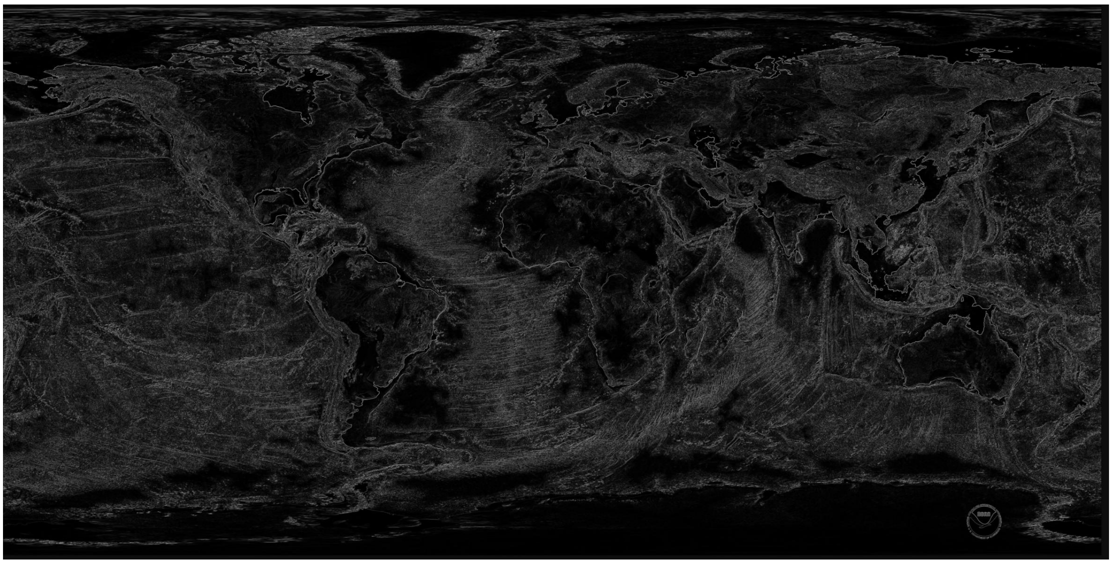
sobel cuda_O3

```
renjianxin@jetson05:~/public_html/test/examples$ nvcc -O3 -o sobel_cuda_O3 sobel_cuda.cu -lfreeimage
renjianxin@jetson05:~/public_html/test/examples$ ls
amdahl_test_mpi.sh       earthmap.jpg                                jetson-hosts.list  sobel_cuda         sobel_mpi_omp.cpp
batchresize.cpp          gustafson_test_mpi.sh                       jetson-nodes.list  sobel_cuda.cu      sobel_ocl.cpp
cathedral_cuda_out.jpg   hello_mpi.cpp                               logutils.h         sobel_cuda_O3      sobel_omp.cpp
cathedral.jpg            IBM_Blue_Gene_P_supercomputer_cuda_out.jpg  oclutils.h         sobel.h
cudautils.h              IBM_Blue_Gene_P_supercomputer.jpg           sobel.cl           sobel_mpi.cpp
earthmap_cuda_out.jpg    imgutils.h                                  sobel.cpp          sobel_mpi_cuda.cu
renjianxin@jetson05:~/public_html/test/examples$
```

```
renjianxin@jetson05:~/public_html/test/examples$ time ./sobel_cuda_03 IBM_Blue_Gene_P_supercomputer.jpg IBM_Blue_Gene_P_supercomputer_cuda_03.jpg
1 CUDA devices found!
CUDA device 0: NVIDIA Tegra X1
  multi processor count: 1
  shared memory per block: 48 KB
  max threads per block: 1024
  max threads per multi processor: 2048
  max warps per multi processor: 64
Use CUDA device 0
the minimum value: 0
the maximum value: 1059.25
The total time for execution is: 0.284049s

real    0m0.599s
user    0m0.188s
sys     0m0.124s
renjianxin@jetson05:~/public_html/test/examples$ time ./sobel_cuda_03 earthmap
```

```
renjianxin@jetson05:~/public_html/test/examples$ time ./sobel_cuda_03 earthmap.jpg earthmap_cuda_03.jpg
1 CUDA devices found!
CUDA device 0: NVIDIA Tegra X1
  multi processor count: 1
  shared memory per block: 48 KB
  max threads per block: 1024
  max threads per multi processor: 2048
  max warps per multi processor: 64
Use CUDA device 0
the minimum value: 0
the maximum value: 962.468
The total time for execution is: 0.54021s

real    0m2.334s
user    0m1.588s
sys     0m0.216s
renjianxin@jetson05:~/public_html/test/examples$ time ./sobel_cuda_03 cathedral.jpg cathedral_cuda_03.jpg
1 CUDA devices found!
CUDA device 0: NVIDIA Tegra X1
  multi processor count: 1
  shared memory per block: 48 KB
  max threads per block: 1024
  max threads per multi processor: 2048
  max warps per multi processor: 64
Use CUDA device 0
the minimum value: 0
the maximum value: 999.015
The total time for execution is: 1.05859s

real    0m4.623s
user    0m3.128s
sys     0m0.536s
renjianxin@jetson05:~/public_html/test/examples$
```





实验结论

	IBM_Blue_Gene_P_supercomputer.jpg			earthmap.jpg			cathedral.jpg		
	time	minimum value	maximum value	time	minimum value	maximum value	time	minimum value	maximum value
sobel serial	7.65837s	0	1052.84	38.3598s	0	942.731	134.356s	0	690.076
sobel serial	1.89975s	0	1052.84	9.44979s	0	942.731	34.1287s	0	690.076

O3									
sobel omp	1.93455s	0	1052.84	9.72089s	0	942.731	33.7722s	0	690.076
sobel omp O3	0.598822s	0	1052.84	2.9676s	0	942.731	10.3436s	0	690.076
sobel cuda	0.171761s	0	1059.25	0.411797s	0	962.468	1.03218s	0	999.015
sobel cuda O3	0.284049s	0	1059.25	0.54021s	0	962.468	1.05859s	0	999.015

实验数据表明，OpenMP 并行化和 -O3 编译器优化都能显著降低 Sobel 边缘检测算法的运行时间。在该硬件平台上，OpenMP 并行算法比串行算法快约 3-4 倍，体现了多核计算在处理图像像素级并行任务时的优势。开启 -O3 优化后，串行算法的性能提升最为显著，大约约 4 倍，说明串行代码对编译器优化非常敏感，此时和并行算法的处理效率接近；并行算法在开启优化后也能获得约 3 倍的性能提升。并行开启-O3 优化后，能获得最快的处理速度，处理 cathedral.jpg 的时间从原始的 134 秒降低到了 10 秒，总性能提升超过 10 倍。

然而，与基于 GPU 的 CUDA 加速方案相比，CPU 层面的优化仍存在数量级上的差距。实验结果显示，CUDA 版本展现出了压倒性的性能优势，其运行速度远超串行及 OpenMP 版本。特别是在处理高分辨率图像时，CUDA 实现了约 100 倍的惊人加速比，即便是在未开启-O3 优化的情况下，其基础性能也已超越了经过极致优化的 CPU 并行代码。这充分证明了在 Sobel 算子这类典型的 SIMD 图像处理任务中，GPU 的大规模并行架构能够有效掩盖数据传输开销，提供比单纯依赖 CPU 多核扩展或编译器指令优化更为强大的算力支持。此外，不同测试图片间的耗时差异也反映出算法性能受图像分辨率及内存访问模式的显著影响，进一步验证了异构计算在处理大规模数据密集型任务时的必要性。

关于-O3 优化选项的影响，实验数据揭示了一个反直觉但合理的现象：该选项对 CPU 和 GPU 代码产生了截然不同的效果。对于 CPU 端的串行及 OpenMP 代码，-O3 带来了显著的性能提升，这得益于 GCC 编译器针对 ARM 架构进行的循环展开及 NEON 向量化指令优化，极大地提高了计算效率，甚至使得优化后的单核串行性能逼近未优化的多核并行性能；然而，对于 GPU 端的 CUDA 代码，强制开启-O3 反而导致运行时间增加。造成这一差异的原因在于 NVCC 编译器自身已具备针对 GPU 寄存器分配和指令调度的复杂优化策略，外部强加的-O3 标志可能干扰了其默认机制或导致设备端代码体积膨胀从而降低缓存命中率，加之 Jetson 平台采用共享内存架构，Sobel 算子属于典型的内存密集型任务，过激的优化可能导致寄存器溢出至本地内存，进而增加了访问延迟并拖慢了整体速度。